VU Quiz Collection

Total Questions: 34 from 34 Quiz(es)

Courses: MTH603

Generated: 1/6/2026, 12:40:16 PM

Powered by Firewall Bypass VU Extension
Waseem Akram | wasii.dev | +1 (226) 798-2360

VU Quiz Collection

Generated by Firewall Bypass VU Extension

Quiz 1 - MTH603 (Q1/10)

Question:

To improve the accuracy of the derivative of the function, which of the following method is more helpful?

Options:

A. Central Difference

B. Interpolation

C. Extrapolation

D. Divided Difference

Copied: 1/6/2026, 10:40:02 AM

Quiz 2 - MTH603 (Q9/10)

Question:

The area of a trapezoid is obtained by adding the area of a ……and a triangle.

Options:

A. None of the given choices

B. Rectangle

C. Circle

D. Square

Copied: 1/6/2026, 10:57:27 AM

Quiz 3 - MTH603 (Q3/10)

Question:

Given the following data

x:1 2 7 11

y:6 10 13 37

Which formula is useful in finding the interpolating polynomial?

Options:

A. Newton’s forward difference interpolation formula

B. None

C. Newton’s backward difference interpolation formula

D. Lagrange’s interpolation formula

Copied: 1/6/2026, 10:40:45 AM

Quiz 4 - MTH603 (Q4/10)

Question:

interpolation

Options:

A. y = f (x) =" 0 . 0 3 8 9 + (x" 1) (2 . 2) + (x" 1) ((x" 4) (0 . 4 3 3 3) y = f (x) =" 0 . 0 3 8 9 + (x" 1) (2 . 2) + 0.0389 + (x - 1)(2.2) + (x - 1)((x - 4)(0.4333)

B. y = f (x) = 2 . 2 + (x" 1) (" 0 . 0 3 8 9) + (x" 1) ((x" 4) (0 . 4 3 3 3) y = f (x) = 2 . 2 + (x" 1) (" 0 . 0 3 8 9) + + (x - 1)(- 0.0389) + (x - 1)((x - 4)(0.4333)

C. y = f (x) = 2 . 2 + (x" 1) (0 . 4 3 3 3) + (x" 1) ((x" 4) (" 0 . 0 3 8 9) y = f (x) = 2 . 2 + (x" 1) (0 . 4 3 3 3) + (+ (x - 1)(0.4333) + (x - 1)((x - 4)(- 0.0389)

D. y = f (x) =" 0 . 0 3 8 9 + (x" 1) (0 . 4 3 3 3) + (x" 1) ((x" 4) (2 . 2) y = f (x) =" 0 . 0 3 8 9 + (x" 1) (0 . 4 3 3 0.0389 + (x - 1)(0.4333) + (x - 1)((x - 4)(2.2)

Copied: 1/6/2026, 10:41:01 AM

Quiz 5 - MTH603 (Q5/10)

Question:

Which of the following is the Cote's number (weighting coefficient) for the function:
f(x) = x+1 in the interval [0,1]?

Options:

A. 3/2

B. -3/2

C. 1/2

D. -1/2

Copied: 1/6/2026, 10:41:58 AM

Quiz 6 - MTH603 (Q6/10)

Question:

points first

Options:

A. none

B. 59

C. 82

D. 76

Copied: 1/6/2026, 10:42:29 AM

Quiz 7 - MTH603 (Q7/10)

Question:

In Simpson's 3/8 rule, the global error is of …….

Options:

A. None of the given choices

B. O(h3)

C. O(h4)

D. O(h2)

Copied: 1/6/2026, 10:43:00 AM

Quiz 8 - MTH603 (Q8/10)

Question:

At which of the following points the Maximum value of 2nd derivative of function f(x) = -(2/x) in the interval:[1,4] exits?

Options:

A. At x=4

B. At x=2

C. At x=3

D. At x=1

Copied: 1/6/2026, 10:43:22 AM

Quiz 9 - MTH603 (Q9/10)

Question:

We prefer ……over the Lagrange's interpolating method for economy of computation.

Options:

A. Newton's forward difference method

B. None of the given choices

C. Newton's backward difference method

D. Newton's divided difference method

Copied: 1/6/2026, 10:43:41 AM

Quiz 10 - MTH603 (Q3/10)

Question:

point interpolation

Options:

A . y = f (x) = (x" x 1) (x" x 2) (x 0" x 1) (x 0" x 2) y 0 + (x" x 0) (x" x 2) (x 1" x 0) (x 1" x 2) y 1 + (x" x 0) (x" x 1) (x" x 2) (x 0" x 1) (x 0" x 2) y 0 + (x" x 0) (x" x 2) (x 1" x 0) (x 1" x 2) y 1 + (x" x 0) (x" x 1) (x 2" - {x_1})(x - {x_2})}}{{({x_0} - {x_1})({x_0} - {x_2})}}{y_0} + \frac{{(x - {x_0})(x - {x_2})}}{{({x_1} - {x_0})({x_1} - {x_2})}}{y_1} + \frac{{(x - {x_0})(x - {x_1})}}{{({x_2} - {x_0})({x_2} - {x_1})}}{y_2}

B . y = f (x) = (x 0" x 1) (x 0" x 2) (x" x 1) (x" x 2) y 0 + (x 1" x 0) (x 1" x 2) (x" x 0) (x" x 2) y 1 + (x 2" x 0) (x 0" x 1) (x 0" x 2) (x" x 1) (x" x 2) y 0 + (x 1" x 0) (x 1" x 2) (x" x 0) (x" x 2) y 1 + (x 2" x 0) (x 2" x 1) (x \frac{{({x_0} - {x_1})({x_0} - {x_2})}}{{(x - {x_1})(x - {x_2})}}{y_0} + \frac{{({x_1} - {x_0})({x_1} - {x_2})}}{{(x - {x_0})(x - {x_2})}}{y_1} + \frac{{({x_2} - {x_0})({x_2} - {x_1})}}{{(x - {x_0})(x - {x_1})}}{y_2}

C . y = f (x) = (x" x 1) (x" x 2) (x 1" x 0) (x 1" x 2) y 0 + (x" x 0) (x" x 2) (x 0" x 1) (x 0" x 2) y 1 + (x" x 0) (x" x 1) (x" x 2) (x 1" x 0) (x 1" x 2) y 0 + (x" x 0) (x" x 2) (x 0" x 1) (x 0" x 2) y 1 + (x" x 0) (x" x 1) (x 2" - {x_1})(x - {x_2})}}{{({x_1} - {x_0})({x_1} - {x_2})}}{y_0} + \frac{{(x - {x_0})(x - {x_2})}}{{({x_0} - {x_1})({x_0} - {x_2})}}{y_1} + \frac{{(x - {x_0})(x - {x_1})}}{{({x_2} - {x_0})({x_2} - {x_1})}}{y_2}

D . y = f (x) = (x" x 1) (x" x 2) (x 0" x 1) (x 0" x 2) y 2 + (x" x 0) (x" x 2) (x 1" x 0) (x 1" x 2) y 1 + (x" x 0) (x" x 1) (x" x 2) (x 0" x 1) (x 0" x 2) y 2 + (x" x 0) (x" x 2) (x 1" x 0) (x 1" x 2) y 1 + (x" x 0) (x" x 1) (x 2" - {x_1})(x - {x_2})}}{{({x_0} - {x_1})({x_0} - {x_2})}}{y_2} + \frac{{(x - {x_0})(x - {x_2})}}{{({x_1} - {x_0})({x_1} - {x_2})}}{y_1} + \frac{{(x - {x_0})(x - {x_1})}}{{({x_2} - {x_0})({x_2} - {x_1})}}{y_0}

Copied: 1/6/2026, 12:36:44 PM

Quiz 11 - MTH603 (Q1/10)

Question:

points

Options:

A. -2

B. -3

C. 2

D. 3

Copied: 1/6/2026, 10:55:04 AM

Quiz 12 - MTH603 (Q6/10)

Question:

points

Options:

A. x 4 8 12y1.31.51.91stD.D0.00620.12ndD.D0.05xy1stD.D2ndD.D 41.30.0062 81.50.10.05 121.9\begin{array}{*{20}{c}}
x&y&{1stD.D}&{2ndD.D} \\\
4&{1.3}&{0.0062}&{} \\\
8&{1.5}&{0.1}&{0.05} \\\
{12}&{1.9}&{}&{}
\end{array}

B. x 4 8 12y1.31.51.91stD.D0.10.352ndD.D0.0062xy1stD.D2ndD.D 41.30.1 81.50.350.0062 121.9\begin{array}{*{20}{c}}
x&y&{1stD.D}&{2ndD.D} \\\
4&{1.3}&{0.1}&{} \\\
8&{1.5}&{0.35}&{0.0062} \\\
{12}&{1.9}&{}&{}
\end{array}

C. x 4 8 12y1.31.51.91stD.D0.10.00622ndD.D0.05xy1stD.D2ndD.D 41.30.1 81.50.00620.05 121.9\begin{array}{*{20}{c}}
x&y&{1stD.D}&{2ndD.D} \\\
4&{1.3}&{0.1}&{} \\\
8&{1.5}&{0.0062}&{0.05} \\\
{12}&{1.9}&{}&{}
\end{array}

D. x 4 8 12y1.31.51.91stD.D0.050.12ndD.D0.0062xy1stD.D2ndD.D 41.30.05 81.50.10.0062 121.9\begin{array}{*{20}{c}}
x&y&{1stD.D}&{2ndD.D} \\\
4&{1.3}&{0.05}&{} \\\
8&{1.5}&{0.1}&{0.0062} \\\
{12}&{1.9}&{}&{}
\end{array}

Copied: 1/6/2026, 12:37:54 PM

Quiz 13 - MTH603 (Q3/10)

Question:

Let f(x,y)=2x3+6y3+9xyf(x,y)=2x3+6y3+9xyf(x,y) = 2{x^3} + 6{y^3} + 9xy For x=0, 1,2,3,4, and y=0, 1,2,3,4 Then computing the value of f(2.5,3.5) is an example of .....

Options:

A. Interpolation in four dimensions

B. Interpolation in two dimensions

C. Interpolation in one dimension

D. Interpolation in three dimensions

Copied: 1/6/2026, 10:55:46 AM

Quiz 14 - MTH603 (Q8/10)

Question:

Which of the following reason(s) lead towards the numerical integration methods?

Options:

A. Integrand is given in tabular form

B. All above choices are true

C. Analytical evaluation of integral is impossible

D. Analytical evaluation of integral is very complicated

Copied: 1/6/2026, 11:15:19 AM

Quiz 15 - MTH603 (Q5/10)

Question:

If the given tabular function f(x) is approximated by the polynomial 'P1(x) = x+1' then which of the following polynomial will approximate the derivative of f(x) ?

Options:

A. 2x

B. x-1

C. x

D. 1

Copied: 1/6/2026, 10:56:18 AM

Quiz 16 - MTH603 (Q6/10)

Question:

points

Options:

A. y[x0,x1,x2]y[x0,x1,x2]y[{x_0},{x_1},{x_2}]

B. y[x0]y[x0]y[{x_0}]

C. y[x0,x1]y[x0,x1]y[{x_0},{x_1}]

D. y[y0,y1,y2]y[y0,y1,y2]y[{y_0},{y_1},{y_2}]

Copied: 1/6/2026, 10:56:37 AM

Quiz 17 - MTH603 (Q7/10)

Question:

points

Options:

A. x 2 4 6y0.311.21stD.D" 0.06250.352ndD.D0.1xy1stD.D2ndD.D 20.3" 0.0625
y}{*{20}{c}}
x&y&{1stD.D}&{2ndD.D} \\\
2&{0.3}&{ - 0.0625}&{} \\\
4&1&{0.35}&{0.1} \\\
6&{1.2}&{}&{}
\end{array}

B. x 2 4 6y0.311.21stD.D0.350.12ndD.D" 0.0625xy1stD.D2ndD.D 20.30.35 41

```
y}{*{20}{c}}
  x&y&{1stD.D}&{2ndD.D} \\\
  2&{0.3}&{0.35}&{} \\\
  4&1&{0.1}&{ - 0.0625} \\\
  6&{1.2}&{}&{}
\end{array}
```

```
C. x 2 4 6y0.311.21stD.D0.1" 0.06252ndD.D0.35xy1stD.D2ndD.D 20.30.1 41" 0.06250.35 61.2\be
 x&y&{1stD.D}&{2ndD.D} \\\
 2&{0.3}&{0.1}&{} \\\
 4&1&{ - 0.0625}&{0.35} \\\
 6&{1.2}&{}&{}
\end{array}
```

```
D. x 2 4 6y0.311.21stD.D" 0.06250.12ndD.D0.35xy1stD.D2ndD.D 20.3" 0.0625 410.10.35 61.2\be
 x&y&{1stD.D}&{2ndD.D} \\\
 2&{0.3}&{ - 0.0625}&{} \\\
 4&1&{0.1}&{0.35} \\\
 6&{1.2}&{}&{}
\end{array}
```

Copied: 1/6/2026, 10:56:53 AM

Quiz 18 - MTH603 (Q8/10)

Question:

interpolation

Options:

A. Newton’s interpolation formula

B. Newton’s backward difference formula

C. Newton’s forward difference formula

D. Lagrange’s interpolation formula

Copied: 1/6/2026, 10:57:11 AM

Quiz 19 - MTH603 (Q10/10)

Question:

In Lagrange’s interpolation, for n values of y corresponding to n values of x, we can represent the function f (x) by a polynomial of degree

Options:

A. n+1

B. n+2

C. n-1

D. n

Copied: 1/6/2026, 10:57:42 AM

Quiz 20 - MTH603 (Q1/10)

Question:

´="""´="""\delta \,\, = \,\, - - - -

Options:

A. None

B. E12+E"12E12+E"12{E^{\frac{1}{2}}}\,\, + \,\,\,{E^{ - \,\,\,\,\frac{1}{2}}}

C. E12+E"122E12+E"122\frac{{{E^{\frac{1}{2}}}\,\, + \,\,\,{E^{ - \,\,\,\,\frac{

D. E12" E"12E12" E"12{E^{\frac{1}{2}}}\,\, - \,\,{E^{ - \,\,\,\,\frac{1}{2}}}

Copied: 1/6/2026, 11:12:38 AM

Quiz 21 - MTH603 (Q8/10)

Question:

points

Options:

A. y[x0,x1]y[x0,x1]y[{x_0},{x_1}]

B. y[y0]y[y0]y[{y_0}]

C. y[x0]y[x0]y[{x_0}]

D. y[y0,y1]y[y0,y1]y[{y_0},{y_1}]

Copied: 1/6/2026, 12:38:52 PM

Quiz 22 - MTH603 (Q3/10)

Question:

In Lagrange's interpolation, for the given five points we can represent the function f (x) by a polynomial of degree

Options:

A. 6

B. 5

C. 4

D. 3

Copied: 1/6/2026, 11:13:34 AM

Quiz 23 - MTH603 (Q4/10)

Question:

In Romberg's method, accuracy of Simpson and Trapezoidal rules is improved by ---------.

Options:

A. extrapolation

B. interpolation

Copied: 1/6/2026, 11:13:50 AM

Quiz 24 - MTH603 (Q5/10)

Question:

1st ordered divided difference formula is defined as

Options:

A. y[x0,x1]=(y1-y0)/(x1+x0)

B. y[x0,x1]=(y1-y0)/(x1-x0)

C. None of the given choices

D. y[x0,x1]=(y1+y0)/(x1-x0)

Copied: 1/6/2026, 11:14:07 AM

Quiz 25 - MTH603 (Q6/10)

Question:

: What will be the value of first order divided difference f[1,5]for the following data

x:0 1 5

y:2 1 5

Options:

A. 1

B. 3

C. 2

D. 0

Copied: 1/6/2026, 11:14:37 AM

Quiz 26 - MTH603 (Q2/10)

Question:

points zero

Options:

A. none

B. 46

C. 35

D. 42

Copied: 1/6/2026, 12:36:21 PM

Quiz 27 - MTH603 (Q9/10)

Question:

To evaluate a definite integral of tabular function f(x), piecewise linear approximation led to ---------.

Options:

A. Romberg's Method

B. Simpson's 3/8 Rule

C. Simpson's 1/3 Rule

D. Trapezoidal Method

Copied: 1/6/2026, 11:15:49 AM

Quiz 28 - MTH603 (Q10/10)

Question:

Simpson's rule is a numerical method that approximates the value of a definite integral by using ……polynomials.

Options:

A. Cubic

B. Quadratic

C. Linear

D. None of the given choices

Copied: 1/6/2026, 11:16:16 AM

Quiz 29 - MTH603 (Q1/10)

Question:

In Newton-Cotes formula for finding the definite integral of a tabular function, which of the following is taken as an approximate function then find the desired integral?

Options:

A. Exponential Function

B. Trigonometric Function

C. Polynomial Function

D. Logarithmic Function

Copied: 1/6/2026, 12:36:03 PM

Quiz 30 - MTH603 (Q4/10)

Question:

Simpson's 3/8 rule represents the area between the curve y = f(x) in the interval say [a,b] above x-axis by approximating the given curve by the ----------.

Options:

A. Cubic curve through four points

B. Cubic curve through two points

C. Cubic curve through three points

D. Cubic curve through one point

Copied: 1/6/2026, 12:37:09 PM

Quiz 31 - MTH603 (Q5/10)

Question:

Zero-th order divided difference is defined as

Options:

A. y[x0]=y0

B. None of the given choices

C. y[x0]=x0

D. y[x0]=y1

Copied: 1/6/2026, 12:37:30 PM

Quiz 32 - MTH603 (Q7/10)

Question:

W h a t w i l l b e t h e v a l u e o f 2 a 2 i n t h e g i v e n d i v i d e d i f f e r e n c e t a b l e ? x 2 4 6 8 y 0 . 5 1 . 1 1 . 7 2 . . 2 d D . D" 0 . 0 0 2 1 W h a t w i l l b e t h e v a l u e o f 2 a 2 i n t h e g i v e n d i v i d e d i f f e r e n c e t a b l e ? x y 1 s t D . D 2 n d D . D 3 r d D . D 2 0 . 5 0 . 3 a 4 1 . 1 0 . 3" 0 . 0 1 2 5" 0 . 0 0 2 1 6 1 . 7 0 . 2 5 8 2 . . 2 W h a t w i l l divide difference table? \ *20c x&y&1stD.D&2ndD.D&3rdD.D 2&0.5&0.3&a& 4&1.1&0.3& - 0.0125& - 0.0021 6&1.7&0.25&& 8&2..2&&&

Options:

A. 0.0893

B. 0.0115

C. 0.0612

D. 0

Copied: 1/6/2026, 12:38:28 PM

Quiz 33 - MTH603 (Q9/10)

Question:

For a function ‘f(x) = x’, with a step size of ‘h=0.01’, which of the following gives the 1st derivative at x =1 by using two point formula?

Options:

A. y’(x) = 1.01 + Some Truncation Error

B. y’(x) = 0.1 + Some Truncation Error

C. y’(x) = -0.1 + Some Truncation Error

D. y’(x) = 1+Some Truncation Error

Copied: 1/6/2026, 12:39:11 PM

Quiz 34 - MTH603 (Q10/10)

Question:

points

Options:

A. 0

B. -1

C. -3

D. -2

Copied: 1/6/2026, 12:39:45 PM
